

Project Plan

StokvelConnect — Stokvel Management Platform

Collective 4

1. Project Overview

StokvelConnect is a web-based stokvel management platform built to digitise and simplify the management of informal savings groups (stokvels) in South Africa. The platform provides role-based dashboards for admins, treasurers, and members, enabling contribution tracking, payout scheduling, meeting management, and financial analytics.

Technology Stack:

- Frontend: Vanilla JavaScript + HTML/CSS
- Backend / Database: Supabase (PostgreSQL + Auth + RLS)
- Authentication: Google OAuth via Supabase
- Payments: PayFast sandbox integration
- Hosting: Microsoft Azure (stokvelconnect.azurewebsites.net)
- CI/CD: GitHub Actions with CodeCov
- Testing: Jest

2. Team Members

#	Full Name	Role	Student No.
1	Mashudu	Developer	
2	Boitumelo	Developer	
3	Emmanuel	Developer	
4	Daniel	Developer	
5	Fhatuwani	Developer	

3. Project Milestones

Milestone	Start Date	Review Date	Key Deliverables
Sprint 1	04 Apr 2026	13 Apr 2026	Project setup, repo structure, initial UI, Supabase config, auth flow
Sprint 2	14 Apr 2026	20 Apr 2026	Group creation, member invites, accept invite flow, UML diagrams
Sprint 3	21 Apr 2026	11 May 2026	Contributions, PayFast integration, meetings, payouts, analytics, burndown charts, CI/CD
Sprint 4	12 May 2026	19 May 2026	Profile editing, notifications, record contribution, disburse modal, payment banner, export PDF, user_roles table, final UML
Final Submission	19 May 2026	22 May 2026	All artefacts compiled, video recorded, Moodle submission

4. Sprint Plan

Sprint 1 — 13 April 2026

Goal: Establish project foundation and core infrastructure.

- Set up GitHub repository and Azure hosting environment
- Configure Supabase project, PostgreSQL schema, and Row Level Security
- Implement Google OAuth login and role-based redirect
- Build base HTML/CSS layout and navigation
- Create initial admin and member dashboard skeletons

Sprint 2 — 14–20 April 2026

Goal: Core group and member management features.

- Group creation form and Supabase insert flow
- Invite members by email with token-based accept flow
- Group listing (my-groups page)
- UML diagrams: Use Case and Class Diagram
- Product backlog refinement and sprint backlog for Sprint 2

Sprint 3 — 21 April – 11 May 2026

Goal: Financial and operational features.

- Contribution tracking with status management (pending/confirmed/flagged)
- PayFast sandbox payment integration
- Meeting scheduling and minutes recording
- Payout schedule with drag-to-order and disbursement
- Analytics dashboard with charts and CSV export
- SARB interest rate fetching and projection calculator
- GitHub Actions CI/CD pipeline with CodeCov
- Sprint burndown charts

Sprint 4 — 12–18 May 2026

Goal: Sprint 4 features and platform polish.

- Profile view and edit (US6)
- Notifications page with clear functionality (US7)
- Record contribution manually (treasurer)
- Disburse modal and payment confirmation banner
- Export analytics as PDF
- user_roles table and role management
- Final UML diagrams (Component + Deployment + updated Sequence diagrams)
- Bug fixes: logout buttons, change-role column removal

5. Risk Management

Risk	Likelihood	Mitigation
Azure deployment downtime	Low	Monitor via Azure portal; redeploy from GitHub Actions if needed
Supabase API limits or outage	Low	RLS policies ensure data safety; Supabase free tier monitored
PayFast sandbox changes	Medium	Using sandbox environment only; no real transactions at risk
Team member unavailability	Medium	Tasks distributed across all 5 members; Git history reflects contributions
Deadline slippage	Low	Weekly sprint reviews with tutor kept team on track

6. Tools & Methodology

Methodology:

- Scrum — 4 sprints with weekly/biweekly tutor review sessions
- Taiga used for product backlog, sprint backlogs, burndown charts, and retrospectives
- Daily stand-up summaries documented per sprint

Development Tools:

- VS Code + Node.js 20
- Git + GitHub (version control and CI/CD)
- GitHub Actions (automated testing and deployment)
- CodeCov (code coverage reporting)
- Jest (unit testing)
- draw.io (UML diagrams)